OUTDATED DOCUMENT — DO NOT CITE OR DEPOSIT.

This LaTeX rendering has not been regenerated since the scientific reframing of July 2026. It still presents the *spectral* account of the dead zone, which was **refuted** on 6 July 2026 (the boundary is governed by coupling degree, not by algebraic connectivity), reports ablation ratios that were deliberately withdrawn on 8 July 2026 (near-zero denominator), and states that the preprint has been submitted — it has **not** been submitted, and version V6.0.0 has **not** been deposited on Zenodo.

The current source of record is `COMPENDIUM.md`, in this same directory. It supersedes this file in full, and additionally reports what the project has *refuted about itself*, which this version omits entirely.

Mem4ristor — Technical Compendium

An executive summary of the model, validation, and neuromorphic hardware mapping.

Julien Chauvin | Independent Researcher | github.com/cafe-virtuel/Mem4ristor | DOI 10.5281/zenodo.18620596 | V6.0.0 — 13 July 2026

1. Context & Methodology

Mem4ristor is an open-science research project exploring the spatiotemporal dynamics of doubt-modulated networks of non-linear oscillators. It addresses a fundamental neuromorphic challenge: preventing consensus-driven dynamic collapse in scale-free topologies.

Developed independently by Julien Chauvin, this project prioritizes strict mathematical validation, reproducibility, and physical hardware feasibility. All code, verification scripts, and raw simulation outputs are version-controlled and public, offering a fully traceable pedigree for all reported findings. The methodology emphasizes rigorous cross-validation by parallel independent peer-review simulations before committing results, ensuring a high standard of scientific integrity.

2. In Numbers

Active development	Since August 2025
Formal investigations	18 key study areas
Simulations run	$> 5{,}000$ runs
Automated unit tests	**130 passing — 0 failures**
Papers	1 preprint submitted (Zenodo) · paper_B in preparation
Zenodo DOI	10.5281/zenodo.18620596
GitHub Repository	cafe-virtuel/Mem4ristor

3. The Model in 3 Paragraphs

Mem4ristor simulates networks of FitzHugh-Nagumo (FHN) neurons — a classical model of neuronal oscillators. The central extension is the **doubt variable** $u \in [0, 1]$: each node carries its

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own level of doubt, which dynamically modulates the *polarity* of its coupling with its neighbours. When $u \approx 0$ or $u \approx 1$, a node becomes "heretical" — it actively pushes against local consensus, preventing the synchronous collapse of the network.

u is not a control parameter. It is a structural mechanism. The proof: locking u at its initial value (FROZEN_U ablation) explodes pairwise synchrony (Kuramoto order parameter R) by $+2326\%$, transforming a functional and diverse network into a perfectly aligned herd. That is not a nuance — it is an order-of-magnitude jump. u is what prevents consensus from devouring diversity.

What the model predicts: a **spectral dead zone** governed by the mean coupling degree of the network. Above a critical coupling degree, no input can reactivate cognitive diversity — *topology alone* locks the network. Below that threshold, chimera states emerge spontaneously: coexistence of partial synchronization and chaos, through a mechanism distinct from all known classical chimeras.

4. The Findings

[1] The Topological Dead Zone
Topology dictates cognitive destiny. Above the mean coupling threshold, no input can reactivate the network. Connectivity kills diversity.

$\rightarrow$ **Complete separation confirmed. 100% accuracy. Formal logistic regression.**
Figure: `fiedler_phase_diagram.png`

[2] u = anti-synchronization filter — the FROZEN_U surge
Locking the doubt variable transforms a functional network into a synchronized herd. u is not a parameter — it is what prevents consensus from devouring diversity.

$\rightarrow$ **Pairwise synchrony goes from 0.031 (FULL) to 0.751 (FROZEN_U). Factor $\times$24.2.**
Figure: `p2_sigma_social_ablation.png`

[3] Topological Intelligence (LZ per node)
In a functional FULL network, hubs have *more structured* trajectories than peripheral nodes. In a frozen FROZEN_U network, the correlation disappears. u couples individual complexity to local connectivity — an emergent property, not anticipated by model design.

$\rightarrow$ $r = -0.716$ **(BA $m = 5$, $N = 400$, $p = 1.29 \times 10^{-79}$). Result not predicted by model design.**
Figure: `lz_per_node.png`

[4] Chimeras — a mechanistically distinct class
Abrams-Strogatz (2004): chimera by fixed non-local coupling (*quenched*). Mem4ristor: chimera by dynamic polarity modulation via $u(t)$, on a scale-free network without imposed symmetry. $R = 0.141$ (deep desynchronization) vs. $R = 0.766$ for AS. Both are chimeras. They are not the same.

$\rightarrow$ R **Mem4ristor** $= 0.141$ **vs.** R **Abrams-Strogatz** $= 0.766$**. Two mechanistically distinct classes.**
Figure: `reviewer2_chimera_comparison.png`

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5. Robustness

Question	Protocol	Result
Real chimera state or pure noise?	Kuramoto R, 3 conditions	$R = 0.513$ FULL vs $R = 0.211$ noise ✓
Does u have a causal role?	Transfer entropy, heretics $\leftrightarrow$ network	TE proven in both directions ✓
Do FULL/FROZEN distributions overlap?	Cohen U3, $n = 50$	U3$= 100\%$ — strictly disjoint ✓
Does spatial noise break the dead zone?	Matérn noise (4 structures), BA $m = 5$	All break it at $\eta = 0.1$ ✓
Is the isolated node stable?	Linear analysis, Jacobian	$v^* = -1.286$, sub-Hopf confirmed ✓
Is the Euler integrator sufficient?	RK45 vs Euler	Max $\Delta(H_{\text{cog}}) < 0.006$ ✓
Results invariant under network size?	Finite-size scaling $N = 100 \to 4000$	Scale-invariant mode validated ✓
Is the sigmoid fine-tuned?	Slope robustness $1.0 \to 10.0$	Stable plateau $H \in [2.8, 3.2]$ ✓
Is the spectral gap stable?	Fiedler $N \to 4000$	Fiedler≈ 2.86 stable ✓
Sensitive to initial conditions?	5 IC types	$H \approx 3.5$ regardless of IC ✓
Travelling wave or chaos?	Max-TLCC temporal lag	Max-TLCC$= 0.410$ — spatiotemporal chaos ✓

6. The Honest Moment

We initially searched for the critical exponents of the spectral transition, yielding $\beta \approx 0$ ($R^2 = -0.002$). The power law did not hold, initially suggesting an abrupt, first-order transition. We documented this negative result transparently.

However, pushed by our own AI "Red Team" to provide definitive thermodynamic proof, we executed a massive finite-size scaling campaign ($N \in \{100, 200, 400, 800, 1600\}$) computing the **Binder cumulant** U_4. This exhaustive "heroic run" (1600×1600 dense matrices) conclusively demonstrated that the transition is a **continuous crossover rather than a thermodynamic phase transition**. The Binder cumulant U_4 remains essentially flat at the Gaussian reference value of $2/3$ across all system sizes, ruling out a critical thermodynamic transition at the boundary.

An honest negative result led to a rigorous methodological correction, ultimately proving the exact nature of the phase transition.

7. The Hardware Bridge & Scaling Budgets

We validated the dead-zone escape in SPICE simulation (ngspice 46) on a BA $m = 5$ network of 64 nodes. While software integration paths strictly follow topological constraints toward structural death, thermal noise inherent to physical analog substrates naturally frustrates the dead zone attractor.

A 50-seed Monte Carlo campaign establishes this escape as a Cohen's $d = 20.78$ effect ($p <$

10^{-63}), restoring continuous entropy to $H \approx 4.30$ bits. In contrast, digital Gaussian perturbation fails to escape the trap even at $270\times$ the equivalent amplitude, highlighting the qualitative computational role of physical thermodynamic fluctuations.

Furthermore, we modeled the physical implementation budgets for a 3000-step trajectory on a network of $N = 100$ nodes:

- **Spintronics (STNO vortex)**: $T_{\text{node}} = 1$ ns, $dt_{\text{phys}} = 2.25$ ps. Physical emulation completes in 6.75 ns and consumes ~ 2.02 nJ (based on a nominal power of 3 mW per vortex).
- **Photonics (GST phase-change)**: $T_{\text{node}} = 100$ ns, $dt_{\text{phys}} = 225$ ps. Emulation completes in 675 ns and consumes ~ 384 fJ (signal-only, at 10 photons per step at 1550 nm).
- **Electrical (NbO_2 Mott Neuristor)**: $T_{\text{node}} = 1$ μs, $dt_{\text{phys}} = 2.25$ ns. Emulation completes in 6.75 μs and consumes ~ 67.5 nJ (assuming 225 fJ per node-step).

Compared to a simulated digital CPU step, physical emulation pathways offer a $10^5\times$ to $10^8\times$ speedup and a $10^9\times$ to $10^{14}\times$ reduction in energy consumption.

8. Reproduce in 5 Minutes

```
git clone https://github.com/cafe-virtuel/Mem4ristor.git
cd Mem4ristor
pip install -r requirements.txt
pip install -e .
python experiments/demo_chimera.py
# → demo_chimera_output.png : time dynamics · phase space · topology
```

The visual validation script generates temporal dynamics (heretic frustration vs. majority), Kuramoto phase space, and BA $m = 3$ topology in a single output plot.

9. Scientific Outlook & Collaboration

This work seeks to establish a concrete bridge between network topology and neuromorphic hardware dynamics. We are actively seeking:

- **Academic Partnerships**: Collaborations to physically manufacture and test these networks on hybrid silicon/memristor or photonic substrates.
- **Scientific Feedback**: Dialogue regarding the underlying mathematical theory of doubt-modulated networks and the topological boundaries of synchronization.

If the model, the mathematical framework, or the hardware scaling budgets align with your research interests, please feel free to contact us.

GitHub: github.com/cafe-virtuel/Mem4ristor | **Zenodo**: 10.5281/zenodo.18620596 | **Contact**: contact@cafevirtuel.org | jul.chauvin@gmail.com | **X**: @Jusyl80

All figures and values reported in this compendium are derived directly from the automated verification suite and code repository. No rounding, no omissions.